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GROUP PROJECT REPORT

THREE-DEPARTMENT ENTERPRISE NETWORK DESIGN

TFB1093: DATA COMMUNICATION AND NETWORK

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1.0 PROJECT OVERVIEW

This report records the design and simulation of a robust, secure, and efficient enterprise network for the XYZ Corporation, a fictional mid-sized enterprise, because it is expanding its operations to Kuala Lumpur, but nevertheless, it has maintained its headquarters in Beijing and branch offices in Shanghai and Chengdu. The project examines the central networking issues that the XYZ Corporation is facing, specifically, poor network segmentation, lack of centralized management of its IP addresses, and less-than-optimal internet access arrangements.

This project aims at designing and simulating an example of a three-department business network (HR, Finance, IT) within Huawei eNSP with the focus on the most important characteristics that lie across the Application, Network, and Transport layers. There are specific tasks to be tested, including the installation of application-layer services, including DHCP, DNS, and HTTP/HTTPS; the enabling of network-level facilities, including subnetting, VLANs, and OSPF routing; as well as the illustration of transport-level protocol behavior, in particular the reliability of TCP and the characteristics of UDP performance. The suggested architecture will allow making inter-office communication secure, internal data transfer processes very efficient, and ensure the stable access of clients to a web-based application, which will solve the existing network-related issues and align with the gradual growth of XYZ Corporation.

2.0 Network Architecture and Topology Design

2.1 Topology Selection Rationale

The design of the XYZ Corporation multi-site enterprise network was implemented using a hybrid topology. This architecture combines the redundancy and direct inter-site connection of a full mesh WAN with the centralized management and economic benefits of a star WAN within each local area network (LAN). The reason behind this arrangement can be rationalized by a set of factors:

- 1) Full Mesh Inter-Site Connectivity (WAN):
 - High Redundancy and Resiliency: All of the routers (Beijing HQ, KL, Shanghai, Chengdu) are directly linked to each other, and it is possible to have more than one independent

path that the data in one router can get to another router. The failure of a single link or router does not cause service interruption because it is possible to re-route traffic at any moment through another direct route, which can reduce downtime to keep businesses functional.

- Direct Communication and Fault Isolation: The inter-branch communications are carried out in the most direct route, minimizing the latency and possibly increasing the throughput.

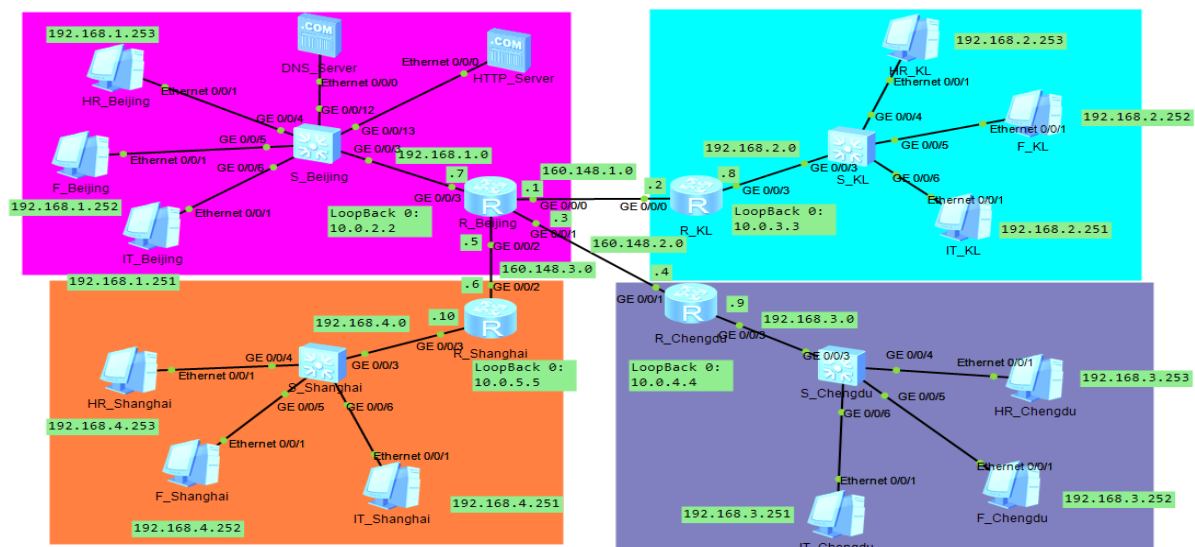
2) Star Intra-Site Connectivity (LAN):

- Scalability: New branch locations are easily added by simply connecting them to the main router. Such flexibility of expansion comes without the need to re-architect existing inter-site links through tricky re-architectural design solutions.
- Cost Effectiveness (initial deployment): A star topology reduces carrying costs compared to a local mesh network because it uses fewer links, and hence the cabling complexity can be minimized. Initial cabling and interface cost will be lower, and can also reduce both installation and long-term maintenance requirements. This will be helpful when the specific enterprise is a mid-sized enterprise that is growing in the number of branches.

The hybrid system combines both a full mesh network and a star topology to form a strong and cost-effective network. The network architecture provides reliability, performance, and simplicity to manage, which makes it a suitable solution for the critical enterprise network of XYZ Corporation.

2.2 Topology Design

The network topology configuration was conceptualized to overcome the operational requirements by having a hybrid architecture, whereby the full mesh structure linking the routers in the central office at Beijing HQ (R Beijing) connects with routers in local offices at Kuala Lumpur (R KL), Shanghai (R Shanghai), and Chengdu (R Chengdu). In each of the branch sites, the internal connectivity needs are handled by a star topology whereby a core router and a complement of switches and workstations handle the departmental segmentation.



3.0 Technology Selection and Implementation Strategy

3.1 Hardware Infrastructure Choices

The virtual hardware given will comprise the enterprise network infrastructure in which the eNSP solution will be simulated.

1) Routers

- Huawei AR series Router (e.g., AR2220 and AR1220) will be used as the core routers at the Beijing HQ router (R_Beijing) and the branch routers Kuala Lumpur (R_KL), Shanghai (R_Shanghai), and Chengdu (R_Chengdu). These devices provide powerful routing capabilities and broad WAN interface coverage needed in operating OSPF and DHCP Relay services, and in inter-VLAN routing.

2) Switches

- Layer 3 Switches
 - The Kuala Lumpur branch (and conceptually each of the other branches/HQ) will feature Huawei S5700 series switches (e.g., S5700) that will support segmenting the switches by individual departments (using VLANs) and allow inter-VLAN routing. These switches have enhanced capabilities such as VLAN, VRP, and routing capabilities.
- Layer 2 Switch
 - Layer 2-enabled Huawei S3700 or S5700 series switches will be installed in every department as an access switch and will provide connectivity to end-user devices.

3) Servers

- PC (as Server)
 - An ordinary PC station located in the Beijing HQ can serve as the HTTP Web Server and host a web service internally accessible over the network or through a simple web server application by eNSP, in case it supports this feature.
- PC (as Server)
 - Another typical PC may be used as a DNS Server in the Beijing HQ location, where it will serve name resolution for its internal domains.

4) End-User Devices

- PCs
 - The PCs will be used to represent the users in the HR, Finance, and IT departments in every branch. They are mainly used to test network connectivity, connect to web services, and demonstrate DHCP operations.
- IP Phones
 - IP phone devices can also be added to the IT department to generate simulated VoIP traffic, but further VoIP configuration is beyond the simple HTTP, DNS, and OSPF eNSP simulation. They are mainly designed to show why real-time traffic prioritization is necessary.

3.2 Network Protocol Implementation

In the same layer, HTTP and HTTPS (Hypertext Transfer Protocol and its secure protocol) are used to achieve communication between client and server via web-based communication. The simplest possible HTTP web server will be installed at the Beijing HQ, and clients at the other branches will be able to reach it in their browsers or via the HTTP commands. As much as HTTPS ensures secure communication, the eNSP simulation will mainly use HTTP to exemplify the simulation. Under network layer architecture, IPv4 subnetting will perform logical division of the network into manageable-sized infra-subnets at the level of individual organizational units, which are HR, Finance, and IT. The action also limits the traffic of broadcasting, strengthens performance, and helps with making the routing secure. This way, every department and every WAN cable will have its IP range allocated and described under a well-organized addressing scheme.

To complement this type of segmentation, VLANs (Virtual Local Area Networks) will be implemented on the Kuala Lumpur site. Distinct VLANs will be created for HR (VLAN 10), Finance (VLAN 20), and IT (VLAN 30), inter-VLAN routing will be activated, and across the department, communication will take place with logical isolation. Every VLAN will receive a different subnet. The dynamic routing protocol we will use over the whole network will be OSPF (Open Shortest Path First). It shall be routed over all routers such as the ones located in Beijing (R_Beijing) and Kuala Lumpur (R_KL), Shanghai (R_Shanghai), and Chengdu (R_Chengdu), with Area 0 being the backbone. OSPF guarantees fast and efficient re-discovery of routes and reconvergence when a route or change takes place. OSPF will also be used to enable authentication in order to protect routing updates. Also, the ICMP (Internet Control Message Protocol) will be utilized in basic diagnostics of the network. Tools like ping and traceroute will help check connection (connectivity) and how to get routing between client devices and server devices.

3.3 IP Addressing Scheme and Subnet Planning

Subnet	Location	Assigned Devices
192.168.1.0/24	Beijing HQ LAN (Servers & PCs)	R-Beijing LAN Interface (192.168.1.254), HTTP Server (192.168.1.252), DNS Server (192.168.1.253)
160.148.2.0/30	WAN Link: Beijing ↔ KL	R-Beijing GE0/0/0 (160.148.2.1), R-KL GE0/0/0 (160.148.2.2)
160.148.3.0/30	WAN Link: Beijing ↔ Shanghai	R-Beijing GE0/0/1 (160.148.3.1), R-Shanghai GE0/0/0 (160.148.3.2)
160.148.4.0/30	WAN Link: Beijing ↔ Chengdu	R-Beijing GE0/0/2 (160.148.4.1), R-Chengdu GE0/0/0 (160.148.4.2)
160.148.5.0/30	WAN Link: KL ↔ Shanghai	R-KL GE0/0/1 (160.148.5.1), R-Shanghai GE0/0/1 (160.148.5.2)
160.148.6.0/30	WAN Link: KL ↔ Chengdu	R-KL GE0/0/2 (160.148.6.1), R-Chengdu GE0/0/1 (160.148.6.2)
160.148.7.0/30	WAN Link: Shanghai ↔ Chengdu	R-Shanghai GE0/0/2 (160.148.7.1), R-Chengdu GE0/0/2 (160.148.7.2)
192.168.10.0/24	KL HR Department (VLAN 10)	KL L3 Switch SVI (192.168.10.254), HR PCs
192.168.20.0/24	KL Finance Department (VLAN 20)	KL L3 Switch SVI (192.168.20.254), Finance PCs

192.168.30.0/24	KL IT Operations (VLAN 30)	KL L3 Switch SVI (192.168.30.254), IT PCs, VoIP Phones
192.168.3.0/24	Shanghai LAN	R-Shanghai LAN Interface (192.168.3.254), Shanghai PCs
192.168.4.0/24	Chengdu LAN	R-Chengdu LAN Interface (192.168.4.254), Chengdu PCs
10.0.0.1/32	Loopback0 (R_Beijing)	R_Beijing Router ID, Management
10.0.0.2/32	Loopback0 (R_KL)	R_KL Router ID, Management
10.0.0.3/32	Loopback0 (R_Shanghai)	R_Shanghai Router ID, Management
10.0.0.4/32	Loopback0 (R_Chengdu)	R_Chengdu Router ID, Management

4.0 Device Configuration and Command Implementation

4.1 Dynamic Host Configuration Protocol (DHCP)

For example, Beijing branch

1.enable dhcp at router and create ip pool

```
R_Beijing
R_Beijing
R_Beijing#dhcp enable
Info: The operation may take a few seconds. Please wait for a moment.done.
R_Beijing
Jul 28 2025 17:08:10-08:00 R_Beijing DS/4/DATASYN_CFOCHANGE:OID 1.3.6.1.4.1.201
1.5.25.191.3.1 configurations have been changed. The current change number is 7,
the change loop count is 0, and the maximum number of records is 4095.
R_Beijing#ip pool pool1
R_Beijing-ip-pool-pool1#network 192.168.1.0 mask 24
R_Beijing-ip-pool-pool1#
Jul 28 2025 17:09:40-08:00 R_Beijing DS/4/DATASYN_CFOCHANGE:OID 1.3.6.1.4.1.201
1.5.25.191.3.1 configurations have been changed. The current change number is 8,
the change loop count is 0, and the maximum number of records is 4095.
R_Beijing-ip-pool-pool1#gateway-list 192.168.1.7
R_Beijing-ip-pool-pool1#
Jul 28 2025 17:10:10-08:00 R_Beijing DS/4/DATASYN_CFOCHANGE:OID 1.3.6.1.4.1.201
1.5.25.191.3.1 configurations have been changed. The current change number is 9,
the change loop count is 0, and the maximum number of records is 4095.
R_Beijing-ip-pool-pool1#lease day 1 hour 12
R_Beijing-ip-pool-pool1#
Jul 28 2025 17:10:20-08:00 R_Beijing DS/4/DATASYN_CFOCHANGE:OID 1.3.6.1.4.1.201
1.5.25.191.3.1 configurations have been changed. The current change number is 10,
the change loop count is 0, and the maximum number of records is 4095.
R_Beijing-ip-pool-pool1#quit
R_Beijing
R_Beijing
R_Beijing#dhcp select global
Error: Unrecognized command found at '' position.
R_Beijing
R_Beijing
R_Beijing#int g0/0/3
R_Beijing-GigabitEthernet0/0/3#dhcp select global
R_Beijing-GigabitEthernet0/0/3#
Jul 28 2025 17:11:20-08:00 R_Beijing DS/4/DATASYN_CFOCHANGE:OID 1.3.6.1.4.1.201
1.5.25.191.3.1 configurations have been changed. The current change number is 11,
the change loop count is 0, and the maximum number of records is 4095.
R_Beijing-GigabitEthernet0/0/3#quit
R_Beijing#quit
<R_Beijing#
```

2. display ip pool before dhcp applied on the switch

```
<R_Beijing#display ip pool name pool1
Pool-name      : pool1
Pool-No        : 0
Lease          : 1 Days 12 Hours 0 Minutes
Domain-name    : -
DNS-server0    : -
NTPS-server0   : -
Netbios-type   : -
Position       : Local      Status      : Unlocked
Gateway-0      : 192.168.1.7
Mask           : 255.255.255.0
VPN instance   : -

=====
Start      End      Total Used Idle(Expired) Conflict Disable
=====
192.168.1.1 192.168.1.254 253    0    253(0)      0      0
=====
```

3. enable DHCP at switch

```
(S_Beijing)dhcp enable
Info: The operation may take a few seconds. Please wait for a moment.done.
(S_Beijing)
Jul 28 2025 17:13:17-08:00 S_Beijing DS/4/DATASYNC_CFGCHANGE:OID 1.3.6.1.4.1.201
1.5.25.191.3.1 configurations have been changed. The current change number is 5,
the change loop count is 0, and the maximum number of records is 4095.
(S_Beijing)interface Vlanif 1
(S_Beijing-Vlanif1)
Jul 28 2025 17:13:31-08:00 S_Beijing \W01PHY/1/PHY(1)(0): GigabitEthernet0/0/
6: change status to down
Jul 28 2025 17:13:32-08:00 S_Beijing \W01PHY/1/PHY(1)(1): GigabitEthernet0/0/
6: change status to up
(S_Beijing-Vlanif1)ip address dhcp-alloc
(S_Beijing-Vlanif1)
Jul 28 2025 17:13:55-08:00 S_Beijing \W01FWNET/4/LINK_STATE(1)(2):The line proto
col IP on the interface Vlanif1 has entered the UP state.
Jul 28 2025 17:13:57-08:00 S_Beijing DS/4/DATASYNC_CFGCHANGE:OID 1.3.6.1.4.1.201
1.5.25.191.3.1 configurations have been changed. The current change number is 7,
the change loop count is 0, and the maximum number of records is 4095.
(S_Beijing-Vlanif1)quit
```

4. display ip pool after applied on both router & switch

```
(R_Beijing)display ip pool name pool1
Pool-name      : pool1
Pool-No        : 0
Lease          : 1 Days 12 Hours 0 Minutes
Domain-name    : -
DNS-server0    : -
NTPS-server0   : -
Netbios-type   : -
Position       : Local      Status      : Unlocked
Gateway-0      : 192.168.1.7
Mask           : 255.255.255.0
VPN instance   : --

=====
      Start      End      Total Used Idle(Expired) Conflict Disable
=====
      192.168.1.1 192.168.1.254 253    1      252(0)      0      0
=====
```

5. Set the DHCP protocol on all PCs and get ip address for each of them

Beijing branch – HR

The screenshot shows the 'HR_Beijing' configuration window with the 'Basic Config' tab selected. The 'Host Name' field is empty. The 'MAC Address' field is set to '54-89-98-E6-69-B0'. Under 'IPv4 Configuration', the 'DHCP' radio button is selected, and the 'Obtain DNS server address automatically' checkbox is checked. The 'IP Address', 'Subnet Mask', and 'Gateway' fields are empty. The 'DNS1' and 'DNS2' fields are also empty. Under 'IPv6 Configuration', the 'Static' radio button is selected, and the 'DHCPv6' radio button is unselected. The 'IPv6 Address' field is set to '::', the 'Prefix Length' is set to '128', and the 'IPv6 Gateway' field is set to '::'. An 'Apply' button is located at the bottom right.

The screenshot shows the 'HR_Beijing' configuration window with the 'Console' tab selected. The console output displays the results of the 'ipconfig' command:

```
Welcome to use PC Simulator!

PC>ipconfig

Link local IPv6 address.....: fe80::5689:98ff:fee6:69b0
IPv6 address.....: :: / 128
IPv6 gateway.....: ::
IPv4 address.....: 192.168.1.253
Subnet mask.....: 255.255.255.0
Gateway.....: 192.168.1.7
Physical address.....: 54-89-98-E6-69-B0
DNS server.....:

PC>|
```

4.2 Open Shortest Path First (OSPF) Routing

Configure OSPF for each router

For example, R_Beijing

```
[R_Beijing]ospf 1 router-id 10.0.2.2
[R_Beijing-ospf-1]area 0
[R_Beijing-ospf-1-area-0.0.0.0]
Jul 28 2025 18:11:33-08:00 R_Beijing DS/4/DATASYNC_CFGCHANGE:OID 1.3.6.1.4.1.201
1.5.25.191.3.1 configurations have been changed. The current change number is 20
, the change loop count is 0, and the maximum number of records is 4095.
[R_Beijing-ospf-1-area-0.0.0.0]network 10.0.2.0 0.0.0.255
[R_Beijing-ospf-1-area-0.0.0.0]
Jul 28 2025 18:11:43-08:00 R_Beijing DS/4/DATASYNC_CFGCHANGE:OID 1.3.6.1.4.1.201
1.5.25.191.3.1 configurations have been changed. The current change number is 21
, the change loop count is 0, and the maximum number of records is 4095.
[R_Beijing-ospf-1-area-0.0.0.0]network 160.148.1.0 0.0.0.255
[R_Beijing-ospf-1-area-0.0.0.0]
[R_Beijing-ospf-1-area-0.0.0.0]network 160.148.2.0 0.0.0.255
[R_Beijing-ospf-1-area-0.0.0.0]
Jul 28 2025 18:12:43-08:00 R_Beijing DS/4/DATASYNC_CFGCHANGE:OID 1.3.6.1.4.1.201
1.5.25.191.3.1 configurations have been changed. The current change number is 23
, the change loop count is 0, and the maximum number of records is 4095.
[R_Beijing-ospf-1-area-0.0.0.0]network 160.148.3.0 0.0.0.255
[R_Beijing-ospf-1-area-0.0.0.0]quit
[R_Beijing-ospf-1]
```

Check OSPF status on each router (display ip routing-table)

R_Beijing

```
<R_Beijing>display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
Destinations : 15      Routes : 15

Destination/Mask    Proto    Pre  Cost   Flags NextHop         Interface
-----
10.0.2.0/24         Direct   0    0       D  10.0.2.2      LoopBack0
10.0.2.2/32         Direct   0    0       D  127.0.0.1     LoopBack0
10.0.3.3/32         OSPF     10   1       D  160.148.1.2   GigabitEthernet
0/0/0
10.0.4.4/32         OSPF     10   1       D  160.148.2.4   GigabitEthernet
0/0/1
10.0.5.5/32         OSPF     10   1       D  160.148.3.6   GigabitEthernet
0/0/2
127.0.0.0/8         Direct   0    0       D  127.0.0.1     InLoopBack0
127.0.0.1/32        Direct   0    0       D  127.0.0.1     InLoopBack0
160.148.1.0/24       Direct   0    0       D  160.148.1.1   GigabitEthernet
0/0/0
160.148.1.1/32      Direct   0    0       D  127.0.0.1     GigabitEthernet
0/0/0
160.148.2.0/24       Direct   0    0       D  160.148.2.3   GigabitEthernet
0/0/1
160.148.2.3/32      Direct   0    0       D  127.0.0.1     GigabitEthernet
0/0/1
160.148.3.0/24       Direct   0    0       D  160.148.3.5   GigabitEthernet
0/0/2
160.148.3.5/32      Direct   0    0       D  127.0.0.1     GigabitEthernet
0/0/2
192.168.1.0/24       Direct   0    0       D  192.168.1.7   GigabitEthernet
0/0/3
192.168.1.7/32      Direct   0    0       D  127.0.0.1     GigabitEthernet
0/0/3
```

4.3 GigabitEthernet Interface Setup

Assign IP address on R_Beijing Router

interface GigabitEthernet 0/0/3 – 192.168.1.7 24

```
[R_Beijing]interface GigabitEthernet 0/0/3
[R_Beijing-GigabitEthernet0/0/3]ip address 192.168.1.7 24
[R_Beijing-GigabitEthernet0/0/3]
Jul 28 2025 16:33:14-08:00 R_Beijing %01IFNET/4/LINK_STATE(1)[0]:The line proto
col IP on the interface GigabitEthernet0/0/3 has entered the UP state.
[R_Beijing-GigabitEthernet0/0/3]quit
```

interface GigabitEthernet 0/0/0 – 160.148.1.1 24

```
[R_Beijing]interface GigabitEthernet 0/0/0
[R_Beijing-GigabitEthernet0/0/0]ip address 160.148.1.1 24
[R_Beijing-GigabitEthernet0/0/0]
Jul 28 2025 16:34:32-08:00 R_Beijing %01IFNET/4/LINK_STATE(1)[1]:The line proto
col IP on the interface GigabitEthernet0/0/0 has entered the UP state.
[R_Beijing-GigabitEthernet0/0/0]quit
```

interface GigabitEthernet 0/0/1 – 160.148.2.3 24

```
[R_Beijing]interface GigabitEthernet 0/0/1
[R_Beijing-GigabitEthernet0/0/1]ip address 160.148.2.3 24
[R_Beijing-GigabitEthernet0/0/1]
Jul 28 2025 16:35:13-08:00 R_Beijing %01IFNET/4/LINK_STATE(1)[2]:The line proto
col IP on the interface GigabitEthernet0/0/1 has entered the UP state.
[R_Beijing-GigabitEthernet0/0/1]quit
```

interface GigabitEthernet 0/0/2 – 160.148.3.5 24

```
[R_Beijing]interface GigabitEthernet 0/0/2
[R_Beijing-GigabitEthernet0/0/2]ip address 160.148.3.5 24
[R_Beijing-GigabitEthernet0/0/2]
Jul 28 2025 16:35:58-08:00 R_Beijing %01IFNET/4/LINK_STATE(1)[3]:The line proto
col IP on the interface GigabitEthernet0/0/2 has entered the UP state.
Jul 28 2025 16:35:59-08:00 R_Beijing DS/4/DATASYNC_CFGCHANGE:OID 1.3.6.1.4.1.201
1.5.25.191.3.1 configurations have been changed. The current change number is 5,
```

4.4 Domain Name System (DNS) Services

DNS_Server

Basic Config | Server Info | Log Info

MAC Address: 54-89-98-7C-0D-F1 (Format:00-01-02-03-04-05)

IPv4 Config

Local Address: 0 . 0 . 0 . 0 Subnet Mask: 255 . 255 . 255 . 0

Gateway: 0 . 0 . 0 . 0 DNS: 0 . 0 . 0 . 0

Ping Test

IPv4 Address: 0 . 0 . 0 . 0 Times: 10

Local State: Device boot ping success: 10 failed: 0

4.5 Virtual LAN (VLAN) Implementation

Implemented when enabling DHCP at switch

```
(S_Beijing)dhcp enable
Info: The operation may take a few seconds. Please wait for a moment.done.
(S_Beijing)
Jul 28 2025 17:13:17-08:00 S_Beijing DS/4/DATASYNC_CFGCHANGE:OID 1.3.6.1.4.1.201
1.5.25.191.3.1 configurations have been changed. The current change number is 5,
the change loop count is 0, and the maximum number of records is 4095.
(S_Beijing)interface Vlanif 1
(S_Beijing-Vlanif1)
Jul 28 2025 17:13:31-08:00 S_Beijing %01PHY/1/PHY(1)(0): GigabitEthernet0/0/
6: change status to down
Jul 28 2025 17:13:32-08:00 S_Beijing %01PHY/1/PHY(1)(1): GigabitEthernet0/0/
6: change status to up
(S_Beijing-Vlanif1)ip address dhcp-alloc
(S_Beijing-Vlanif1)
Jul 28 2025 17:13:55-08:00 S_Beijing %01IPNET/4/LINK_STATE(1)(2):The line proto
col IP on the interface Vlanif1 has entered the UP state.
Jul 28 2025 17:13:57-08:00 S_Beijing DS/4/DATASYNC_CFGCHANGE:OID 1.3.6.1.4.1.201
1.5.25.191.3.1 configurations have been changed. The current change number is 7,
the change loop count is 0, and the maximum number of records is 4095.
(S_Beijing-Vlanif1)quit
```

4.6 Loopback Interface Configuration

For example, LoopBack R_Beijing

```
<R_Beijing>system-view
Enter system view, return user view with Ctrl+Z.
[R_Beijing]interface LoopBack 0
[R_Beijing-LoopBack0]
Jul 28 2025 17:52:12-08:00 R_Beijing DS/4/DATASYNC_CFGCHANGE:OID 1.3.6.1.4.1.201
1.5.25.191.3.1 configurations have been changed. The current change number is 12
, the change loop count is 0, and the maximum number of records is 4096.
[R_Beijing-LoopBack0]ip address 10.0.2.2 24
[R_Beijing-LoopBack0]quit
[R_Beijing]
```

5.0 Network Protocol Behaviour Analysis

5.1 Transport Layer Protocol Comparison (TCP vs UDP)

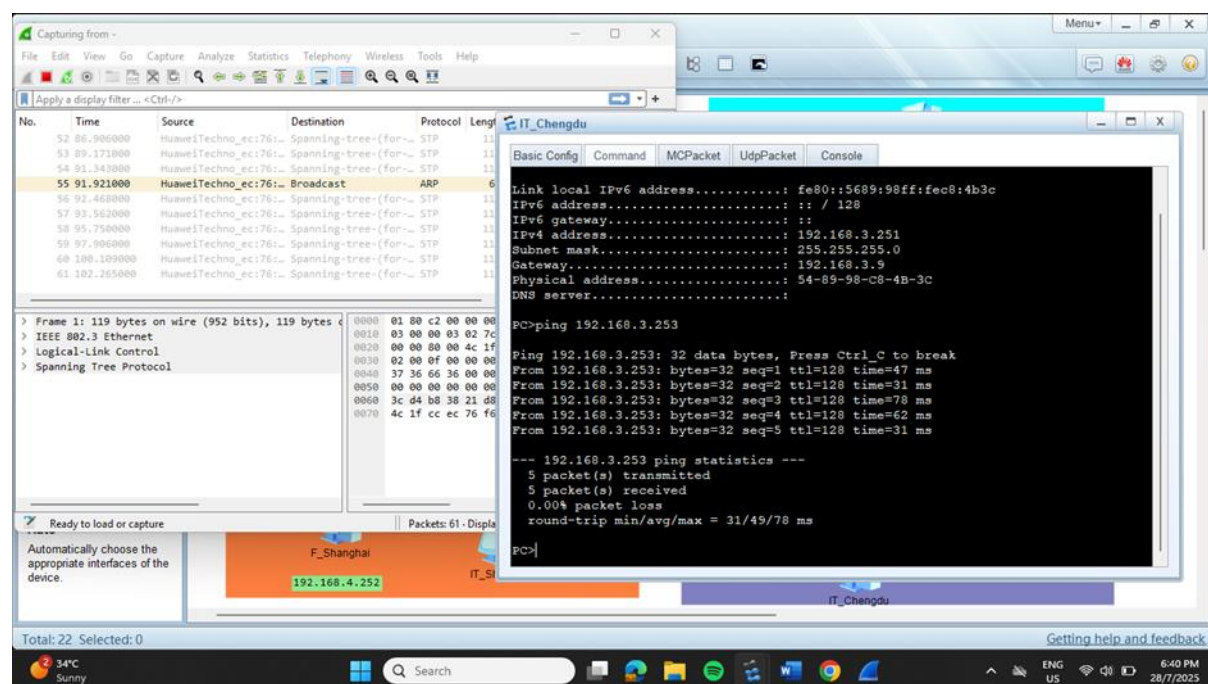
Transmission Control Protocol (TCP) refers to a connection-oriented and dependable protocol that is meant to deliver data completely, non-stop, and with proper sequencing. Before transferring the data, TCP uses a three-way handshaking mechanism to enable an end-to-end connection (SYN, SYN-ACK, ACK). A distinct identifier is to be used by each data packet, which permits its reconstruction when received. The acknowledgment is sent back to the sender as soon as received (ACK) and in case it is not received the transmission unit is sent in case it is not received. Besides, TCP has mechanisms for flow control in order to avoid overloading the slower receivers by a faster source and a congestion control mechanism as a means of overcoming instances of network congestion.

The TCP used at the XYZ Corporation is used in areas where accuracy and security of data is of the foremost importance. It lies behind HTTP/HTTPS and safe commerce through the web browsing experience, it secures file sharing, and protects human resources secrets. Under the finance department, TCP will ensure that proper financial transactions are carried out and recorded. User Datagram Protocol (UDP) is another high-speed, connectionless protocol, which does not consider constructing a preliminary connection and sending acknowledgements and retransmit packets received under the protocol. UDP, therefore, offers low latency and less overhead as compared to TCP, but at the expense of reliability. Information can be lost or repeated, or come backward.

UDP is not used directly in the current infrastructure, but it is still applicable when fast or precise timing is more important than uncompromised accuracy. In the case of XYZ Corporation, the VoIP (Voice over IP) is used in the IT sector with such restrictions. Information loss can be acceptable in voice communication as long as the latency between conversations remains low. Similarly, UDP enables device monitoring, where the application of discussing constant state changes is desirable instead of inflexible verification of each packet arrival. To sum up, TCP is still the best protocol to use where reliability and security of data transfer are crucial, including web access, file transfer, and contact with sensitive information. Although the UDP is not presently in the network of the XYZ Corporation, its existing usage in real-time applications such as VoIP provides the perfect evidence that the adoption of a specific protocol can always be done in line with the functional needs of a given application.

5.2 Traffic Analysis Using Wireshark Packet Capture

Wireshark testing (capture data from IT_Chengdu to HR_Chengdu)



Observation

The research carried out on network simulation feasibility examines network simulation capability in terms of using Wireshark and the built-in network packet capture capacity of emulation tools like Network Simulation Platform (eNSP). It aimed to confirm the working

capacity of protocols specifically, HTTP and ICMP, and to investigate the analysed traffic volume and packet format. Even though ping tests showed that there was connectivity, Wireshark scanned only a few of the expected IP-level activity, rather, it sampled a lot of link-layer events (e.g., STP broadcasts and ARP requests).

Reasons for Missing Traffic

There are a number of reasons that can explain this difference. To start with, the seizure can have been done on a port that was not moving the targeted traffic. Second, no traffic filters are used, and it may blur important packets in the background noise. Third, although ping was active, there is a chance that no HTTP traffic was sent within the timeframe of the capture or that the server was either not active or not responsive. Finally, eNSP creates emulation constraints that can define expected behaviour compared with a real network and may exclude types of traffic in analytical calculations.

TCP (Expected Outputs)

Potentially, HTTP GET requests on the part of the client and the subsequent HTTP responses on the part of the server would follow TCP three-way handshake (SYN, SYN-ACK, ACK) under the optimal conditions. ACK packets would be used to acknowledge the reception of data during an exchange and duplication of ACK packets and retransmission when there was packet loss would provide a demonstration of congestion-control methods provided by TCP.

UDP (Expected Outputs)

Upon investigation of UDP traffic, e.g. VoIP, you might find Wireshark showing the datagrams of a temporal sequence excluded of TCP handshakes and retransmissions. This UDP sparse packet structure is indicative of the uniqueness of UDP that sacrifices reliability assurances in favor of throughput.

5.3 Network Connectivity Verification Testing

Status connection from router (display interface GigabitEthernet x/x/x)

R_Beijing with interface GigabitEthernet 0/0/3

```
(R_Beijing)display int g0/0/3
GigabitEthernet0/0/3 current state : UP
Line protocol current state : UP
Last line protocol up time : 2025-07-28 16:33:14 UTC+08:00
Description:
Route Port,The Maximum Transmit Unit is 1500
Internet Address is 192.168.1.7/24
IP Sending Frames' Format is PKTFMT_ETHNT_2, Hardware address is 5489-9842-0ad6
Last physical up time : 2025-07-28 16:26:21 UTC+08:00
Last physical down time : 2025-07-28 16:26:19 UTC+08:00
Current system time: 2025-07-28 16:49:19+08:00
Hardware address is 5489-9842-0ad6
  Last 300 seconds input rate 53 bytes/sec, 0 packets/sec
  Last 300 seconds output rate 0 bytes/sec, 0 packets/sec
    Input: 74494 bytes, 626 packets
    Output: 0 bytes, 0 packets
  Input:
    Unicast: 0 packets, Multicast: 626 packets
    Broadcast: 0 packets
  Output:
    Unicast: 0 packets, Multicast: 0 packets
    Broadcast: 0 packets
  Input bandwidth utilization :  0%
  Output bandwidth utilization :  0%
```

R_Beijing with interface GigabitEthernet 0/0/0

```
(R_Beijing)display int g0/0/0
GigabitEthernet0/0/0 current state : UP
Line protocol current state : UP
Last line protocol up time : 2025-07-28 16:34:32 UTC+08:00
Description:
Route Port,The Maximum Transmit Unit is 1500
Internet Address is 160.148.1.1/24
IP Sending Frames' Format is PKTFMT_ETHNT_2, Hardware address is 5489-9842-0ad3
Last physical up time : 2025-07-28 16:26:22 UTC+08:00
Last physical down time : 2025-07-28 16:26:19 UTC+08:00
Current system time: 2025-07-28 16:49:42+08:00
Hardware address is 5489-9842-0ad3
  Last 300 seconds input rate 1 bytes/sec, 0 packets/sec
  Last 300 seconds output rate 1 bytes/sec, 0 packets/sec
    Input: 1040 bytes, 11 packets
    Output: 1040 bytes, 11 packets
  Input:
    Unicast: 11 packets, Multicast: 0 packets
    Broadcast: 0 packets
  Output:
    Unicast: 10 packets, Multicast: 0 packets
    Broadcast: 1 packets
  Input bandwidth utilization :  0%
  Output bandwidth utilization :  0%
```

R_Beijing with interface GigabitEthernet 0/0/1

```
[R_Beijing]display int g0/0/1
GigabitEthernet0/0/1 current state : UP
Line protocol current state : UP
Last line protocol up time : 2025-07-28 16:35:13 UTC+08:00
Description:
Route Port,The Maximum Transmit Unit is 1500
Internet Address is 160.148.2.3/24
IP Sending Frames' Format is PKTFMT_ETHNT_2, Hardware address is 5489-9842-0ad4
Last physical up time : 2025-07-28 16:26:21 UTC+08:00
Last physical down time : 2025-07-28 16:26:19 UTC+08:00
Current system time: 2025-07-28 16:50:18+08:00
Hardware address is 5489-9842-0ad4
  Last 300 seconds input rate 1 bytes/sec, 0 packets/sec
  Last 300 seconds output rate 1 bytes/sec, 0 packets/sec
    Input: 1040 bytes, 11 packets
    Output: 1040 bytes, 11 packets
      Input:
        Unicast: 11 packets, Multicast: 0 packets
        Broadcast: 0 packets
      Output:
        Unicast: 10 packets, Multicast: 0 packets
        Broadcast: 1 packets
    Input bandwidth utilization : 0%
    Output bandwidth utilization : 0%
```

R_Beijing with interface GigabitEthernet 0/0/2

```
[R_Beijing]display int g0/0/2
GigabitEthernet0/0/2 current state : UP
Line protocol current state : UP
Last line protocol up time : 2025-07-28 16:35:58 UTC+08:00
Description:
Route Port,The Maximum Transmit Unit is 1500
Internet Address is 160.148.3.5/24
IP Sending Frames' Format is PKTFMT_ETHNT_2, Hardware address is 5489-9842-0ad5
Last physical up time : 2025-07-28 16:26:21 UTC+08:00
Last physical down time : 2025-07-28 16:26:19 UTC+08:00
Current system time: 2025-07-28 16:50:32+08:00
Hardware address is 5489-9842-0ad5
  Last 300 seconds input rate 1 bytes/sec, 0 packets/sec
  Last 300 seconds output rate 1 bytes/sec, 0 packets/sec
    Input: 1040 bytes, 11 packets
    Output: 1040 bytes, 11 packets
      Input:
        Unicast: 11 packets, Multicast: 0 packets
        Broadcast: 0 packets
      Output:
        Unicast: 10 packets, Multicast: 0 packets
        Broadcast: 1 packets
    Input bandwidth utilization : 0%
    Output bandwidth utilization : 0%
```

Status of network (from switch - display interface brief & from router – display ip interface brief)

From S_Beijing

```
<S_Beijing>display interface brief
PHY: Physical
'down: administratively down
(l): loopback
(s): spoofing
(b): BFD down
(e): ETHOAM down
(d1): DLDP down
(d): Dampening Suppressed
InUti/OutUti: input utility/output utility
Interface      PHY    Protocol InUti OutUti  inErrors  outErrors
GigabitEthernet0/0/1    down  down      0%  0%      0         0
GigabitEthernet0/0/2    down  down      0%  0%      0         0
GigabitEthernet0/0/3    up     up        0%  0%      0         0
GigabitEthernet0/0/4    up     up        0%  0%      0         0
GigabitEthernet0/0/5    up     up        0%  0%      0         0
GigabitEthernet0/0/6    up     up        0%  0%      0         0
GigabitEthernet0/0/7    down  down      0%  0%      0         0
GigabitEthernet0/0/8    down  down      0%  0%      0         0
GigabitEthernet0/0/9    down  down      0%  0%      0         0
GigabitEthernet0/0/10   down  down      0%  0%      0         0
GigabitEthernet0/0/11   down  down      0%  0%      0         0
GigabitEthernet0/0/12   up     up        0%  0%      0         0
GigabitEthernet0/0/13   up     up        0%  0%      0         0
GigabitEthernet0/0/14   down  down      0%  0%      0         0
GigabitEthernet0/0/15   down  down      0%  0%      0         0
GigabitEthernet0/0/16   down  down      0%  0%      0         0
GigabitEthernet0/0/17   down  down      0%  0%      0         0
GigabitEthernet0/0/18   down  down      0%  0%      0         0
GigabitEthernet0/0/19   down  down      0%  0%      0         0
GigabitEthernet0/0/20   down  down      0%  0%      0         0
GigabitEthernet0/0/21   down  down      0%  0%      0         0
GigabitEthernet0/0/22   down  down      0%  0%      0         0
GigabitEthernet0/0/23   down  down      0%  0%      0         0
GigabitEthernet0/0/24   down  down      0%  0%      0         0
Meth0/0/1            down  down      0%  0%      0         0
NULL0                up     up(s)     0%  0%      0         0
Vlanif1              up     down      --  --      0         0
```

From R_Beijing

```
<R_Beijing>display ip interface brief
'down: administratively down
'down: FIB overload down
'down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
The number of interface that is UP in Physical is 5
The number of interface that is DOWN in Physical is 6
The number of interface that is UP in Protocol is 5
The number of interface that is DOWN in Protocol is 6

Interface      IP Address/Mask  Physical  Protocol
Ethernet0/0/0   unassigned       down      down
Ethernet0/0/1   unassigned       down      down
GigabitEthernet0/0/0  160.148.1.1/24  up        up
GigabitEthernet0/0/1  160.148.2.3/24  up        up
GigabitEthernet0/0/2  160.148.3.5/24  up        up
GigabitEthernet0/0/3  192.168.1.7/24  up        up
NULL0          unassigned       up        up(s)
Serial0/0/0     unassigned       down      down
Serial0/0/1     unassigned       down      down
Serial0/0/2     unassigned       down      down
Serial0/0/3     unassigned       down      down
```

6.0 System Testing and Performance Evaluation

Test connection

ping 160.148.1.2 from R_Beijing

```
<R_Beijing>ping 160.148.1.2
PING 160.148.1.2: 56 data bytes, press CTRL_C to break
Reply from 160.148.1.2: bytes=56 Sequence=1 ttl=255 time=70 ms
Reply from 160.148.1.2: bytes=56 Sequence=2 ttl=255 time=40 ms
Reply from 160.148.1.2: bytes=56 Sequence=3 ttl=255 time=1 ms
Reply from 160.148.1.2: bytes=56 Sequence=4 ttl=255 time=40 ms
Reply from 160.148.1.2: bytes=56 Sequence=5 ttl=255 time=30 ms

--- 160.148.1.2 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 1/40/70 ms
```

ping 160.148.2.4 from R_Beijing

```
<R_Beijing>ping 160.148.2.4
PING 160.148.2.4: 56 data bytes, press CTRL_C to break
Reply from 160.148.2.4: bytes=56 Sequence=1 ttl=255 time=60 ms
Reply from 160.148.2.4: bytes=56 Sequence=2 ttl=255 time=40 ms
Reply from 160.148.2.4: bytes=56 Sequence=3 ttl=255 time=60 ms
Reply from 160.148.2.4: bytes=56 Sequence=4 ttl=255 time=90 ms
Reply from 160.148.2.4: bytes=56 Sequence=5 ttl=255 time=30 ms

--- 160.148.2.4 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 30/56/90 ms
```

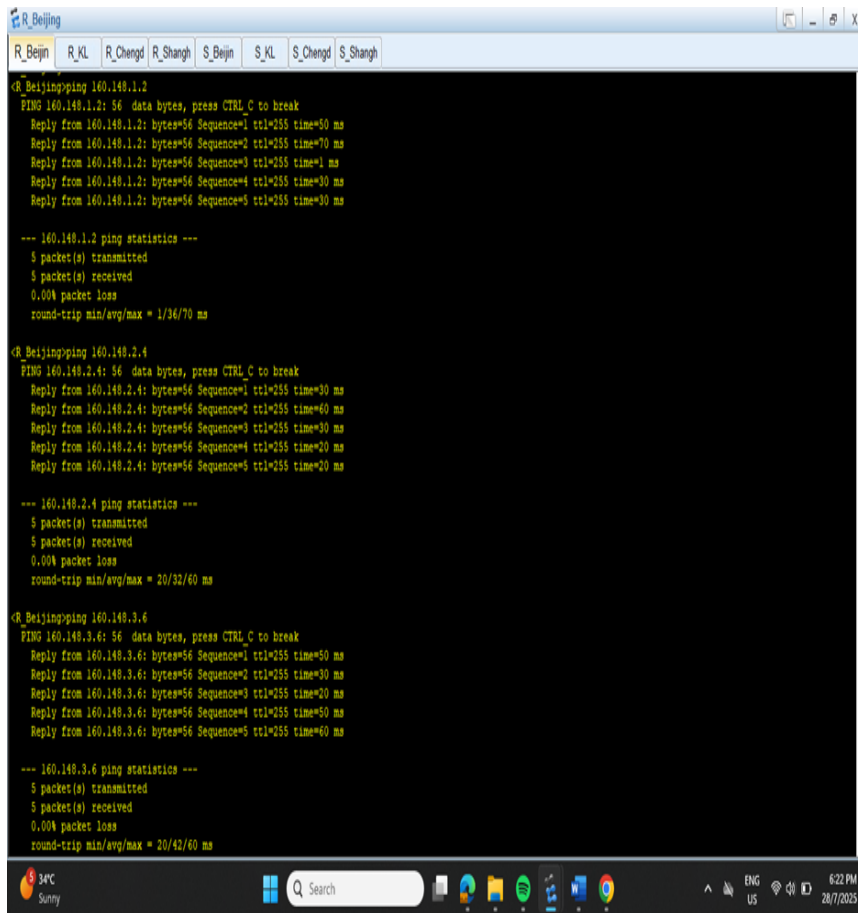
ping 160.148.3.6 from R_Beijing

```
<R_Beijing>ping 160.148.3.6
PING 160.148.3.6: 56 data bytes, press CTRL_C to break
Reply from 160.148.3.6: bytes=56 Sequence=1 ttl=255 time=110 ms
Reply from 160.148.3.6: bytes=56 Sequence=2 ttl=255 time=30 ms
Reply from 160.148.3.6: bytes=56 Sequence=3 ttl=255 time=90 ms
Reply from 160.148.3.6: bytes=56 Sequence=4 ttl=255 time=10 ms
Reply from 160.148.3.6: bytes=56 Sequence=5 ttl=255 time=40 ms

--- 160.148.3.6 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 10/56/110 ms
```

Ping router to check connection after configuring OSPF

R_Beijing



```
R_Beijing
R_Beijing R_KL R_Chengd R_Shangh S_Beijing S_KL S_Chengd S_Shangh
>R_Beijingping 160.148.1.2
PING 160.148.1.2: 56 data bytes, press CTRL_C to break
    Reply from 160.148.1.2: bytes=56 Sequence=1 ttl=255 time=50 ms
    Reply from 160.148.1.2: bytes=56 Sequence=2 ttl=255 time=70 ms
    Reply from 160.148.1.2: bytes=56 Sequence=3 ttl=255 time=1 ms
    Reply from 160.148.1.2: bytes=56 Sequence=4 ttl=255 time=30 ms
    Reply from 160.148.1.2: bytes=56 Sequence=5 ttl=255 time=30 ms

--- 160.148.1.2 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 1/36/70 ms

>R_Beijingping 160.148.2.4
PING 160.148.2.4: 56 data bytes, press CTRL_C to break
    Reply from 160.148.2.4: bytes=56 Sequence=1 ttl=255 time=30 ms
    Reply from 160.148.2.4: bytes=56 Sequence=2 ttl=255 time=60 ms
    Reply from 160.148.2.4: bytes=56 Sequence=3 ttl=255 time=30 ms
    Reply from 160.148.2.4: bytes=56 Sequence=4 ttl=255 time=20 ms
    Reply from 160.148.2.4: bytes=56 Sequence=5 ttl=255 time=20 ms

--- 160.148.2.4 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 20/32/60 ms

>R_Beijingping 160.148.3.6
PING 160.148.3.6: 56 data bytes, press CTRL_C to break
    Reply from 160.148.3.6: bytes=56 Sequence=1 ttl=255 time=50 ms
    Reply from 160.148.3.6: bytes=56 Sequence=2 ttl=255 time=30 ms
    Reply from 160.148.3.6: bytes=56 Sequence=3 ttl=255 time=20 ms
    Reply from 160.148.3.6: bytes=56 Sequence=4 ttl=255 time=50 ms
    Reply from 160.148.3.6: bytes=56 Sequence=5 ttl=255 time=60 ms

--- 160.148.3.6 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 20/42/60 ms
```

7.0 Project Summary and Future Recommendations

Summary

This research project proved that it was successful to design a network architecture and shift the XYZ Corporation using the Huawei eNSP platform. The network, which is aimed at serving three different departments, alleviates several serious networking issues and makes smooth organisational growth possible. It used a Hybrid topology with a Full Mesh WAN configuration used to connect the Beijing headquarters with remote sites continuously, trading on robust and viable connectivity, and Star topologies are used locally at each site. At the Kuala Lumpur location, the Virtual LANs (VLANs) of HR, Finance, and IT were defined and had their unique IP subnets so that the security of the network and the reduction of redundant traffic were promoted. Address assignments were undertaken using the Dynamic Host Configuration Protocol (DHCP), which encourages efficiency in operations and the prevention of any conflict of address allocation.

At the routing level, the Open Shortest Path First (OSPF) protocol was implemented as an inter-router protocol to deliver smart routing decision-making and authentication of high security. All the routers made use of a loopback interface to have a stable router ID. On the application layer, a simple HTTP-based web server and Domain Name System (DNS) were configured at the Beijing headquarters to allow users of all branches access to internal web-based content by resolving domain names. The functions of the transport layer were also described. It was chosen that Transmission Control Protocol (TCP) would be used in reliable data transfer, like internet browsing and file transfer, and User Datagram Protocol (UDP) would be used to utilize fast (connectionless) applications like Voice over IP (VoIP). The use of diagnostic tools was made, such as the ping, traceroute, and packet inspections, to be able to check connectivity, routing, and the performance of the protocols.

To conclude, the simulated network is resilient and secure, scalable, and efficient. It facilitates both internal communication, file sharing, such as client-server applications, and web-based use across all organisational locations, and helps a smooth development of XYZ Corporation.

Recommendations

The suggested measures of improving the network infrastructure of XYZ Corporation include installing security and performance upgrades. First, the installation of firewalls in every location is required to protect the network and to allow VPN connections. Correspondingly, the Network Access Control (NAC) will similarly be utilized to implement so that only authorized devices will be able to access the system. Security auditing will also promote advanced vulnerability detection and remediation periodically. To maximise network resources, Quality of Service (QoS) must be set to give the mission-critical traffic, such as voice calls and financial transactions, priority processing. The backup routers and servers will ensure there is continuity of the network even in case there is a failure of equipment.

Other suggestions are on the need to have a centralised network monitoring and management platform. Both IPv4 and IPv6 are considered advisable to be adopted to be ready for any expected deliberations and future demands. The use of secure Wi-Fi in all offices within the company will enhance support for mobile user activities. Lastly, a backup and disaster recovery solution should be implemented, which will protect important information even when the system is not working. The combination of all these actions is supposed to make the network stronger in terms of resilience, reliability, and ability to accommodate the upcoming needs.

8.0 References

Forouzan, B. A. (2007). *Data communications and networking* (4th ed.). McGraw-Hill Education.

Kurose, J. F., & Ross, K. W. (2017). *Computer networking: A top-down approach* (7th ed.). Pearson.

Peer Evaluation Form

Course: Data Communications and Networks
Project Title: Group Project Report
Group Number: Y
Your Name (Evaluator): Nur Fathiah Aisha
Date: 29/7/2025

Instructions:

Rate each of your group members (excluding yourself) based on their performance and contribution to the project using the criteria below. Be honest and fair. Your feedback will be used as part of the individual assessment.

Group Members Name	Technical Contribution (0-10)	Participation in Meeting (0-5)	Timeliness & Responsibility (0-5)	Teamwork & Communication (0-5)	Total (0-25)
Nur Afina Madihah Binti Muhammad Mazrin	10	5	5	5	25
Nur Husna Syaza Binti Zainal Abidin	10	5	5	5	25
Eiashia Caliana Binti Zurayusmady	10	5	5	5	25
Nur Shalili Amira Binti Badrul Hisham	10	5	5	5	25

Criteria Descriptions:

- Technical Contribution (0–10): Involvement in configuration, simulation, troubleshooting, and documentation.
- Participation in Meetings (0–5): Attended group discussions and contributed meaningfully.
- Timeliness & Responsibility (0–5): Completed assigned tasks on time and took initiative.
- Teamwork & Communication (0–5): Respectful, collaborative, and communicated ideas clearly.

Peer Evaluation Form

Course: Data Communications and Networks
Project Title: Group Project Report
Group Number: Y
Your Name (Evaluator): Nur Afina Madihah
Date: 29/7/2025

Instructions:

Rate each of your group members (excluding yourself) based on their performance and contribution to the project using the criteria below. Be honest and fair. Your feedback will be used as part of the individual assessment.

Group Members Name	Technical Contribution (0-10)	Participation in Meeting (0-5)	Timeliness & Responsibility (0-5)	Teamwork & Communication (0-5)	Total (0-25)
Nur Fathiah Aisha binti Ramniza	10	5	5	5	25
Nur Husna Syaza Binti Zainal Abidin	10	5	5	5	25
Eiashia Caliana Binti Zurayusmady	10	5	5	5	25
Nur Shalili Amira Binti Badrul Hisham	10	5	5	5	25

Criteria Descriptions:

- Technical Contribution (0–10): Involvement in configuration, simulation, troubleshooting, and documentation.
- Participation in Meetings (0–5): Attended group discussions and contributed meaningfully.
- Timeliness & Responsibility (0–5): Completed assigned tasks on time and took initiative.
- Teamwork & Communication (0–5): Respectful, collaborative, and communicated ideas clearly.

Peer Evaluation Form

Course: Data Communications and Networks
 Project Title: Group Project Report
 Group Number: Y
 Your Name (Evaluator): Nur Husna Syaza
 Date: 29/7/2025

Instructions:

Rate each of your group members (excluding yourself) based on their performance and contribution to the project using the criteria below. Be honest and fair. Your feedback will be used as part of the individual assessment.

Group Members Name	Technical Contribution (0-10)	Participation in Meeting (0-5)	Timeliness & Responsibility (0-5)	Teamwork & Communication (0-5)	Total (0-25)
Nur Afina Madihah Binti Muhammad Mazrin	10	5	5	5	25
Nur Fathiah Aisha binti Ramniza	10	5	5	5	25

Eiashia Caliana Binti Zurayusmady	10	5	5	5	25
Nur Shalili Amira Binti Badrul Hisham	10	5	5	5	25

Criteria Descriptions:

- Technical Contribution (0–10): Involvement in configuration, simulation, troubleshooting, and documentation.
- Participation in Meetings (0–5): Attended group discussions and contributed meaningfully.
- Timeliness & Responsibility (0–5): Completed assigned tasks on time and took initiative.
- Teamwork & Communication (0–5): Respectful, collaborative, and communicated ideas clearly.

Peer Evaluation Form

Course: Data Communications and Networks
 Project Title: Group Project Report
 Group Number: Y
 Your Name (Evaluator): Eiashia Caliana
 Date: 29/7/2025

Instructions:

Rate each of your group members (excluding yourself) based on their performance and contribution to the project using the criteria below. Be honest and fair. Your feedback will be used as part of the individual assessment.

Group Members Name	Technical Contribution (0-10)	Participation in Meeting (0-5)	Timeliness & Responsibility (0-5)	Teamwork & Communication (0-5)	Total (0-25)
Nur Afina Madihah Binti Muhammad Mazrin	10	5	5	5	25
Nur Husna Syaza Binti Zainal Abidin	10	5	5	5	25
Nur Fathiah Aisha binti Ramniza	10	5	5	5	25
Nur Shalili Amira Binti Badrul Hisham	10	5	5	5	25

Criteria Descriptions:

- Technical Contribution (0–10): Involvement in configuration, simulation, troubleshooting, and documentation.
- Participation in Meetings (0–5): Attended group discussions and contributed meaningfully.
- Timeliness & Responsibility (0–5): Completed assigned tasks on time and took initiative.

- Teamwork & Communication (0–5): Respectful, collaborative, and communicated ideas clearly.

Peer Evaluation Form

Course: Data Communications and Networks

Project Title: Group Project Report

Group Number: Y

Your Name (Evaluator): Nur Shalili Amira

Date: 29/7/2025

Instructions:

Rate each of your group members (excluding yourself) based on their performance and contribution to the project using the criteria below. Be honest and fair. Your feedback will be used as part of the individual assessment.

Group Members Name	Technical Contribution (0-10)	Participation in Meeting (0-5)	Timeliness & Responsibility (0-5)	Teamwork & Communication (0-5)	Total (0-25)
Nur Afina Madihah Binti Muhammad Mazrin	10	5	5	5	25
Nur Husna Syaza Binti Zainal Abidin	10	5	5	5	25
Eiashia Caliana Binti Zurayusmady	10	5	5	5	25
Nur Fathiah Aisha binti Ramniza	10	5	5	5	25

Criteria Descriptions:

- Technical Contribution (0–10): Involvement in configuration, simulation, troubleshooting, and documentation.
- Participation in Meetings (0–5): Attended group discussions and contributed meaningfully.
- Timeliness & Responsibility (0–5): Completed assigned tasks on time and took initiative.
- Teamwork & Communication (0–5): Respectful, collaborative, and communicated ideas clearly.